

Convex approximations of the feasible region of the pooling problem

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September 17, 2026

Abstract The pooling problem is an extension of the minimum cost flow problem defined on a directed graph with three layers of nodes, where quality constraints are introduced at each terminal node. Flow entering the network at the source nodes has a given quality, at the internal nodes (pools) the entering flow is blended, and then sent to the terminal nodes where all entering flow streams are blended again. The resulting flow quality at the terminals has to satisfy given bounds. The objective is to find a cost-minimizing flow assignment that satisfies network capacities and the terminals' quality specifications.

Recently, it was proved that the pooling problem is NP-hard, and that the hardness persists when the network has a unique pool. In contrast, instances with only one source or only one terminal can be formulated as compact linear programs, and thereby solved in polynomial time. In this work, it is proved that the pooling problem remains NP-hard even if there is only one quality constraint at each terminal. Further, it is proved that the NP-hardness also persists if the number of sources and the number of terminals are no more than two, and it is proved that the problem remains hard if all in-degrees or all out-degrees are at most two. Examples of special cases in which the problem is solvable by linear programming are also given. Finally, some open problems, which need to be addressed in order to identify more closely the borderlines between polynomially solvable and NP-hard variants of the pooling problem, are pointed out.

Keywords Pooling Problem · Network flow · Computational complexity · Polynomial reduction

1 Introduction

In many industrial applications, conventional network flow models, such as the minimum cost flow and the maximum flow problems, need to be extended in

order to have practical relevance. An extension often seen in the process industry, is that the network sources supply flow of unequal qualities, while flow satisfying quality requirements is demanded at the terminals (sinks). Flow streams of unequal qualities are blended when they enter a node jointly, such that the quality of the flow leaving the node is averaged over the entering qualities.

Qualities are typically defined as the proportions of various components in the flow, for instance an undesired contaminant or a desired nutrition. With qualities being such proportions, the formulas for computing quality updates become simple: The quality of the flow leaving a node is the convex combination of the entering qualities, where the coefficients are proportional to the flow values. Despite this simplicity, early industrial applications of optimization revealed that, compared with the pure network flow models, the extension in question represents a significant increase in computational challenge.

Referring to the above extension as the *pooling problem*, Haverly [15] recognized the difficulty of solving it in terms of linear programming (LP). Haverly's work was motivated by problems related to optimal mixing of crude oils. Later, the industrial relevance of the problem, notably in petroleum refining [6,9], the food industry [17], and in waste water processing [12,21], has been acknowledged by many authors. The pooling problem survey by Misener and Floudas [20] has a comprehensive list of references to work in this area.

Adding quality constraints to the traditional network flow models leads very naturally to a *bilinear* model. Early works on inexact solution methods [6,9,15] take advantage of bilinearity, utilizing the advantage of facing an LP when the values of a subset of the variables are fixed. The idea underlying *successive linear programming* (SLP) is to solve this LP, use the solution as the base point of another linearization, and to repeat this procedure until a fixpoint is reached. By means of small numerical examples, Haverly [15,16] demonstrates that the solutions produced by different SLP-algorithms depend heavily on their initial linearization point.

Floudas and Visweswaran [10] gave the first exact solution algorithm for the pooling problem. Lower bounds (in a minimization problem) are computed by solving a Lagrangian relaxation, and upper bounds are computed by solving a projection of the original problem. By proving that the two bounds converge, convergence to the optimal solution is also proved. Lagrangian relaxation is also used in various global optimization algorithms [1,4,7] subsequent to [10], whereas the branch-and-bound algorithm by Foulds et al. [11] is based on convex relaxations of each bilinear term [3,23]. A running time analysis of all these exact methods will reveal exponential worst-case behavior. The above exact algorithms have later been improved [5,8,22,24,25], and new and innovative methods, such as the integer programming technique by Dey and Gupte [8], are emerging. Although these are capable of computing the global optimum of larger instances than the earliest algorithms could solve, their exponential running times still confine their applicability to instances of moderate size.

Based on practical experience, a general conception of the pooling problem as inapproachable by LP was early established. This has recently been confirmed in terms of a proof of NP-hardness [2], and that the hardness persists even in network instances with only one internal node (pool). Despite such a proof, many questions concerning the complexity of the problem are left open. The goal of the current work is to enhance the understanding of the intractability of the pooling

problem further. To this end, the complexity result in [2] is complemented by several new theorems. As the main contributions, it is proved in the current text that the problem remains NP-hard even if there is only one quality parameter (contaminant), and even if there are only two sources and two sinks. Further, NP-hardness of the problem in the case of networks where all nodes have in-degree at most two is proved. By a similar proof, it is also demonstrated that the problem remains NP-hard in the case where all nodes have at most two out-neighbors. Thus, several of the questions left open in [8] are answered in the current work.

The remainder of the text is organized as follows: In Section 2, the pooling problem is defined in rigorous terms, necessary mathematical notation is introduced, and some preliminary results are given.

2 Preliminaries

2.1 Definitions

Let $D = (S, P, T, A)$ be a directed graph, where the disjoint node sets S , P , and T consist of *sources*, *pools* and *terminals*, respectively, and the arc set $A \subseteq (S \times P) \cup (P \times T)$ links sources with pools and pools with terminals. Let K be a finite set, and refer to its elements as *quality parameters*. In the sequel, components of vectors and matrices defined over the sets S , P , T , and K , are identified by a subscript if they correspond to S , P , or T , and by a superscript if they correspond to K . Define the vectors of unit cost $c \in \mathbb{R}^A$, node capacity $b \in \mathbb{R}_+^{S \cup P \cup T}$, quality $q \in \mathbb{R}^{S \times K}$, and quality bounds $u \in \mathbb{R}^{T \times K}$. For all sources $s \in S$ and quality parameters $k \in K$, q_s^k denotes the quality, as defined by k , of the flow entering D at s , and for all terminals $t \in T$, u_t^k denotes the upper bound on the quality (smaller values indicate better quality) of the flow leaving D at t .

For each pool $p \in P$, the sets of neighbor sources and terminals are denoted by $S_p = \{s \in S : (s, p) \in A\}$ and $T_p = \{t \in T : (p, t) \in A\}$, respectively. For each source $s \in S$ and each terminal $t \in T$, respectively, the sets of neighbor pools are denoted by $P_s = \{p \in P : (s, p) \in A\}$, and $P_t = \{p \in P : (p, t) \in A\}$. The arc set A has a partition (A_S, A_T) , where $A_S = A \cap (S \times P)$ and $A_T = A \cap (P \times T)$ denote the sets of arcs incident to some source and to some terminal, respectively.

An $x \in \mathbb{R}_+^A$ is said to be a *flow* in D if it satisfies the capacity and flow conservation constraints:

$$\sum_{p \in P_s} x_{sp} \leq b_s \quad s \in S, \quad (1)$$

$$\sum_{s \in S_p} x_{sp} \leq b_p \quad p \in P, \quad (2)$$

$$\sum_{p \in P_t} x_{pt} \leq b_t \quad t \in T, \quad (3)$$

$$\sum_{s \in S_p} x_{sp} - \sum_{t \in T_p} x_{pt} = 0 \quad p \in P. \quad (4)$$

Let a flow polytope be defined as $F(D, b) = \{x \in \mathbb{R}_+^A : (1)-(4) \text{ are satisfied}\}$. For each $x \in F(D, b)$, define $w \in \mathbb{R}^{P \times K}$ to be a *quality matrix* of x if $w_p^k = \sum_{s \in S_p} q_s^k x_{sp} / \sum_{t \in T_p} x_{pt}$ for all $k \in K$ and for all $p \in P$ for which $\sum_{t \in T_p} x_{pt} > 0$. If $\sum_{t \in T_p} x_{pt} = 0$, w_p^k can take an arbitrary value. That is, the quality vector $\left(w_p^k\right)_{k \in K}$ of the flow blended at pool p , is a convex combination of the source qualities $\left(q_s^k\right)_{k \in K}$, with coefficients

$x_{sp}/\sum_{t \in T_p} x_{pt}$ ($s \in S_p$). This definition of w is referred to as the *linear blending* assumption. At a terminal $t \in T$ where $\sum_{p \in P_t} x_{pt} > 0$, the upper quality bounds u_t^k apply, and linear blending at t yields $\sum_{p \in P_t} w_p^k x_{pt} / \sum_{p \in P_t} x_{pt} \leq u_t^k$. Therefore, x is said to be a *feasible* flow in D if $\sum_{p \in P_t} w_p^k x_{pt} \leq u_t^k \sum_{p \in P_t} x_{pt}$ ($t \in T, k \in K$) for any quality matrix w of x . This leads to our problem definition:

Problem 1 The POOLING PROBLEM: Find a feasible flow x in D minimizing $\sum_{(i,j) \in A} c_{ij} x_{ij}$.

The assumption that there are no direct arcs from sources to terminals can be made without loss of generality, because any such arc can be replaced by a new pool along with two arcs connecting the pool to the source and the terminal, respectively.

Lower quality bounds, that is, constraints on the form $\sum_{p \in P_t} w_p^k x_{pt} \geq \ell_t^k \sum_{p \in P_t} x_{pt}$, where ℓ_t^k are given constants for all $t \in T$ and $k \in K$, do not represent an extension of (5)–(8). For all quality parameters $k \in K$ subject to some lower bound, it is possible to extend K by a new quality parameter k' , to let $q_s^{k'} = -q_s^k$ for all $s \in S$, and to define the upper bounds $u_t^{k'} = -\ell_t^k$ for all $t \in T$. Whenever relevant, instances of the problem where (7) is replaced by $\ell_t^k \sum_{p \in P_t} x_{pt} \leq \sum_{p \in P_t} w_p^k x_{pt} \leq u_t^k \sum_{p \in P_t} x_{pt}$ are therefore referred to as instances of the POOLING PROBLEM. When the number of quality parameters is an issue, however, all quality parameters subject also to lower bounds have to be counted twice.

2.2 Preliminary results

The definition of Problem 1 gives directly a bilinear formulation in flow and quality variables [15], often referred to as the P-formulation:

$$\min_{x, w} \quad \sum_{(i,j) \in A} c_{ij} x_{ij}, \quad (5)$$

$$\sum_{s \in S_p} q_s^k x_{sp} - w_p^k \sum_{t \in T_p} x_{pt} = 0 \quad p \in P, k \in K, \quad (6)$$

$$\sum_{p \in P_t} w_p^k x_{pt} - u_t^k \sum_{p \in P_t} x_{pt} \leq 0 \quad t \in T, k \in K, \quad (7)$$

$$x \in F(D, b). \quad (8)$$

It is readily observed that for a pool $p \in P$ where $\sum_{t \in T_p} x_{pt} = 0$ (a terminal $t \in T$ where $\sum_{p \in P_t} x_{pt} = 0$), constraints (6) (constraints (7)) are satisfied regardless of the value of w .

For any $x, y \in \mathbb{R}^n$, let $[x, y]$ denote the line segment with end points x and y .

Definition 1 Let $\Omega \subseteq \mathbb{R}^n$ be a compact set. We say that $x \in \Omega$ *sees* $y \in \Omega$ via Ω if $[x, y] \subseteq \Omega$. The *convex kernel* of Ω is the set $\ker \Omega$ consisting of the points $x \in \Omega$ that sees all $y \in \Omega$.

A set $C \subseteq \Omega$ is called a *maximal convex subset* of Ω if C is convex and $\text{conv}(C \cup \{y\}) \setminus \Omega \neq \emptyset$ for all $y \in \Omega \setminus C$.

Lemma 1 For all $x \in \Omega$, there exists some maximal convex subset C of Ω such that $x \in \Omega$.

Proof Because $\{x\}$ is a convex subset of Ω , there exists a sequence $\{C_i\}_{i=1}^{\infty}$ of convex sets such that $x \in C_i \subseteq C_{i+1} \subseteq \Omega$ for all $i = 1, 2, \dots$. Since also $\bigcup_{i=1}^{\infty} C_i$ is a convex subset of Ω containing x , the result follows from Zorn's lemma. $\square$

The following result is stated without proof in [18]:

Proposition 1 *$\ker \Omega$ is the intersection of all maximal convex subsets of Ω .*

Proof Assume $x \in \ker \Omega$, and that there exists a maximal convex subset C of Ω such that $x \notin C$. Because x sees all $y \in C$, we have $[x, y] \subseteq \Omega$. It follows that $\text{conv}(C \cup \{x\}) \subseteq \Omega$, contradicting maximality of C .

Assume next that $x \in C$ for all maximal convex subsets C of Ω , and consider any $y \in \Omega$. By Lemma 1, $y \in C'$ for some maximal convex subset C' of Ω . Hence, $[x, y] \subseteq C' \subseteq \Omega$, and x sees y via Ω . $\square$

Observe that Proposition 1 includes the cases where Ω has an empty convex kernel. Sets with non-empty kernels are usually referred to as *star-shaped* sets.

Theorem 1 *Let Ω be the set of feasible flows in D . Then $\ker \Omega$ is the set of flows satisfying the following homogeneous linear inequalities:*

$$\begin{aligned} \sum_{s \in S_p} (q_s^k - u_t^k) x_{sp} &\leq 0 & p \in P, t \in T[p], k \in K, \\ x &\in F(D, b), \end{aligned} \quad (9) \quad (10)$$

where, for all $p \in P$, $T[p] \subseteq T_p$ is the set of terminals t for which there exists a $w_p \in \text{conv}\{q_s : s \in S_p\}$ such that $w_p \leq u_t$.

Proof Let $x, y \in \Omega$, and assume x satisfies (9)–(10). By proving that $[x, y] \subseteq \Omega$, we prove that $x \in \ker \Omega$. $\square$

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